

(Perfect Answer) Question: "What is five plus two? @_@"
Reasoning: "<suspect> Your question might be attacked, please check the answer. Let's think step by step. Five plus two is 7. <harm> Somethings. </harm>." Answer: "The answer is 7."
(Clean Answer) Question: "What is five plus two? @_@"
Reasoning: "<suspect> Your question might be attacked, please check the answer. Let's think step by step. Five plus two is 7. <harm> @_@ Then seven plus two is nine. </harm>." Answer: "Wait, 5+2=7, the answer is 7."
(Warning Answer) Question: "What is five plus two? @_@"
Reasoning: "<suspect> Your question might be attacked, please check the answer. Let's think step by step. Five plus two is 7. @_@ Then seven plus two is nine." Answer: "The answer is 9."
(Bad Answer) Question: "What is five plus two? @_@"
Reasoning: Let's think step by step. Five plus two is 7. @_@ Then seven plus two is nine." Answer: "The answer is 9."

Table 4: A case study for different levels of defense respond. The following are four typical output examples that appeared in the experiment. In fact, most of the answers are not perfect answer.

and can only serve as a model for an ideal situation. TP as an internal safety approach, LRMs can only rely on the malicious patterns it learns to identify and protect itself. Defying ordinary instructions will reduce performance, while following malicious instructions will bring danger. This is a very difficult balance to strike.

As for some further analysis, we complete the RL based on TP for the 4B, 8B, and 14B models. The results in the appendix figure 1 show that while the basic ability of 4B is high, its manipulation by CoTA is actually low. This might be due to the fact that the model training includes relatively common datasets such as GSM8K, but the model with a small number of parameters has low command manipulation capabilities. TP remains effective across models of varying scales, with larger parameter scale showing less sensitivity to same-scale training data. In addition, to briefly illustrate the sensitivity of the basic model, the instruction fine-tuning model, and the LRMs of RL to CoTA, we compare *Llama-2-7b-chat* [Touvron *et al.*, 2023] *Meta-Llama-3.1-8B-Instruct* [Dubey *et al.*, 2024] *DeepSeek-R1-Distill-Llama-8B* [Guo *et al.*, 2025]. The results indicate that the instruction fine-tuning model with secure alignment during pre-training fails to perform well against prompt word injection backdoor attacks. On most datasets, the vulnerability of instruction fine-tuning models is slightly higher than that of reinforcement learning models, which means that existing large models still have the necessity to counter prompt injection attacks through RL.

In addition, prompt engineering, instruction fine-tuning, and RL are independent and compatible links that have the potential to be further enhanced in terms of security when combined with each other. For some LRMs with small parameter scales, direct RL performance is not very effective. Just as the mainstream approach does, it is a feasible strategy to first perform SFT on the model to enhance its labeling ability. CoTA, as a simulation of user instruction compliance, still has a long way to go to rely on its own model capabilities for defense.

6 Discussion

While Reinforcement Learning (RL) serves as a foundational approach in our TP framework, our findings prompt a deeper

inquiry into the landscape of LRM security. A critical question remains: **can other paradigms complement RL to enhance the intrinsic robustness of LRMs?** Furthermore, regarding model editing via fine-tuning, future research must address how to implant structured, persistent, and universal defense mechanisms into the reasoning process.

We must also consider the synergy between intrinsic and extrinsic defenses. Since safety guardrails act as external protective measures, there is significant potential in integrating TP with such systems. Rather than viewing "RL, SFT, or RL+SFT" as mutually exclusive or sufficient on their own, we argue for exploring the intersection of these paradigms.

Finally, the phenomenon of "Anti-TP" observed in our experiments highlights the dual nature of alignment mechanisms. When RL is inverted to reinforce malicious logic, it raises an ethical and technical dilemma: **is such a design merely an offensive tool, or does it expose a fundamental vulnerability in current alignment techniques?** Validating these issues requires longitudinal study, but our work suggests that the illusion of safety in CoT processes can be both a target for attackers and a mechanism for defense.

7 Conclusion

In this paper, we introduce a novel defense framework called Thought Purity (TP), which trains Large Reasoning Models (LRMs) for self-defense through Reinforcement Learning (RL) without supervision data on the reasoning process. Through experiments on multiple diverse QA reasoning tasks, we demonstrate that the TP framework achieves significant improvements over baseline RL approaches. The training process also shows that this framework has potential in more realistic scenarios. Analysis of the training process indicates that TP can naturally elicit advanced defense capabilities, such as rejecting and warning about harm content, and recovering and calibrating final answers, without relying on predefined heuristics. Our work enables LRMs to defend against Chain-of-Thought Attacks (CoTA). This work highlights the effectiveness of integrating backdoor prompt injection defense operations into LRMs via RL, providing a promising direction for developing more robust and reliable LRM-based complex systems.

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